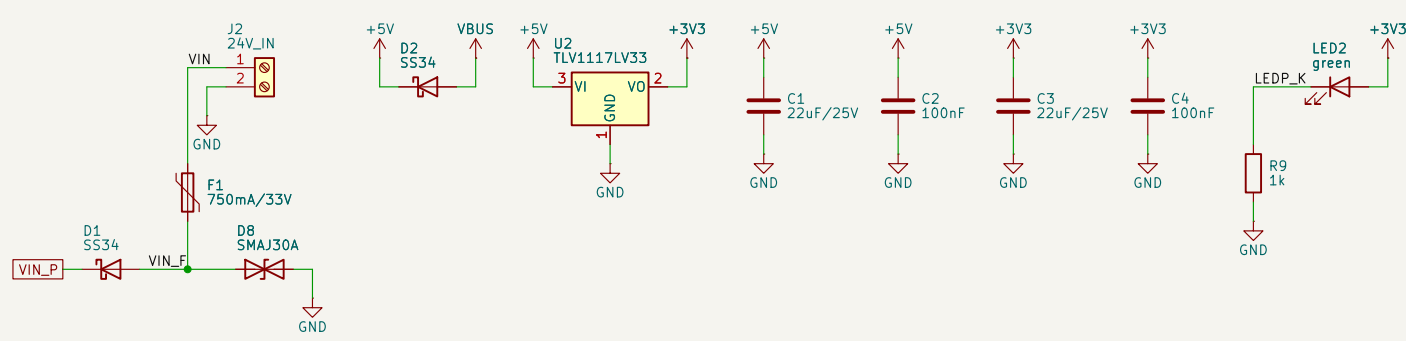
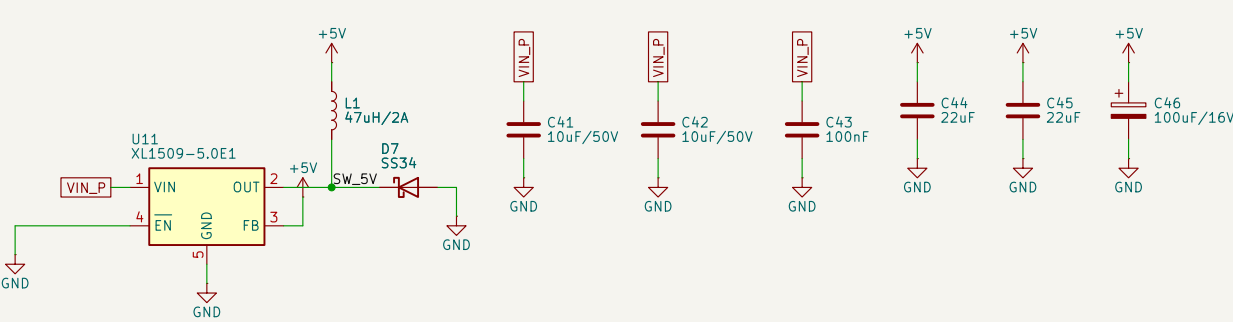


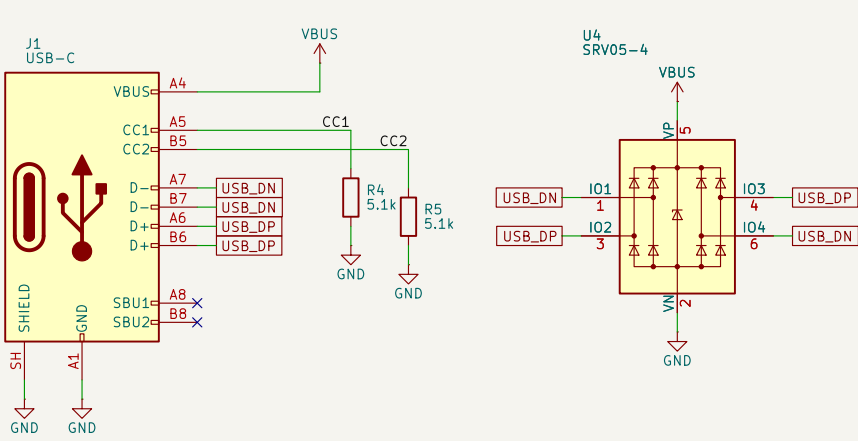
POWER IN: 24 VDC terminal or USB
F1+D8 clamp-and-blow at the entry: D1 blocks reverse polarity.
U11 (right) makes +5V; D2 ORs in USB VBUS; U2 drops it to +3V3



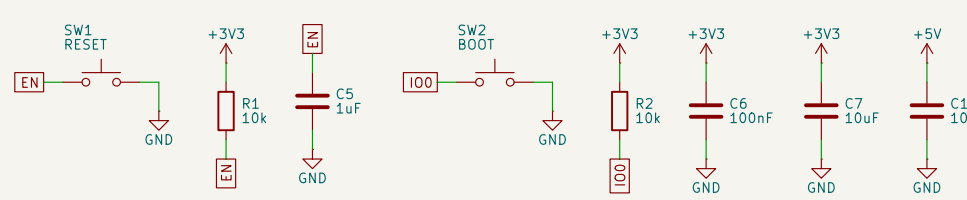
24 V -> 5 V BUCK XL1509-5.0, fixed output
C41-C43 in, D7 catch, L1 47uH, C44/C45/C46 out. FB ties to the output because the part is the fixed 5.0 V version; EN is active LOW and grounded. C46 is an ELECTROLYTIC on purpose: voltage-mode with fixed compensation needs its ESR zero, and an all-ceramic output has none. Keep C41/C42 - U11 - D7 tight: that loop, not L1, is what radiates.



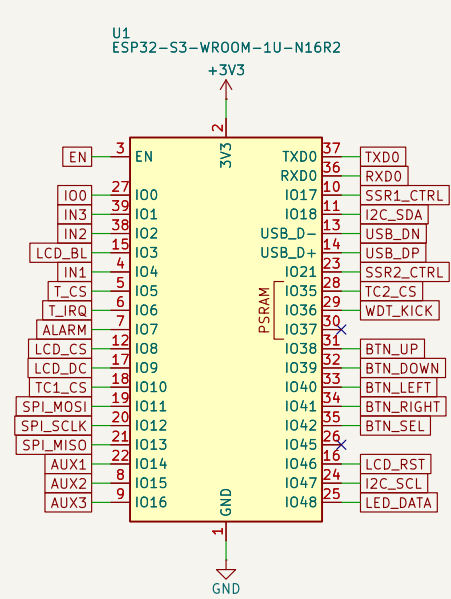
USB-C
native USB flashing + ESD



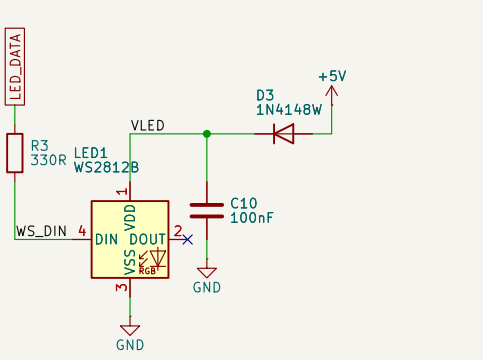
RESET / BOOT / DECOUPLING



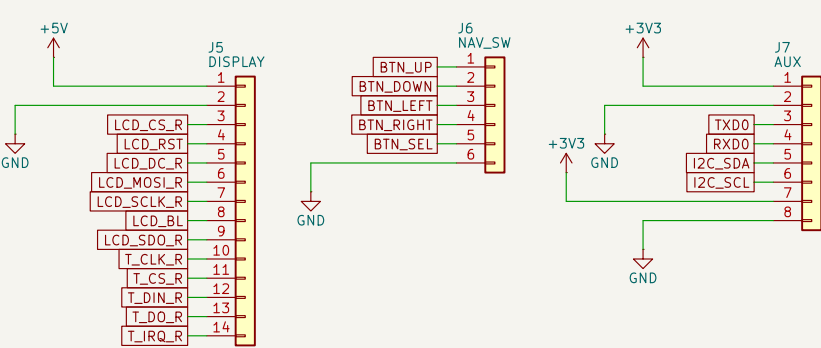
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults



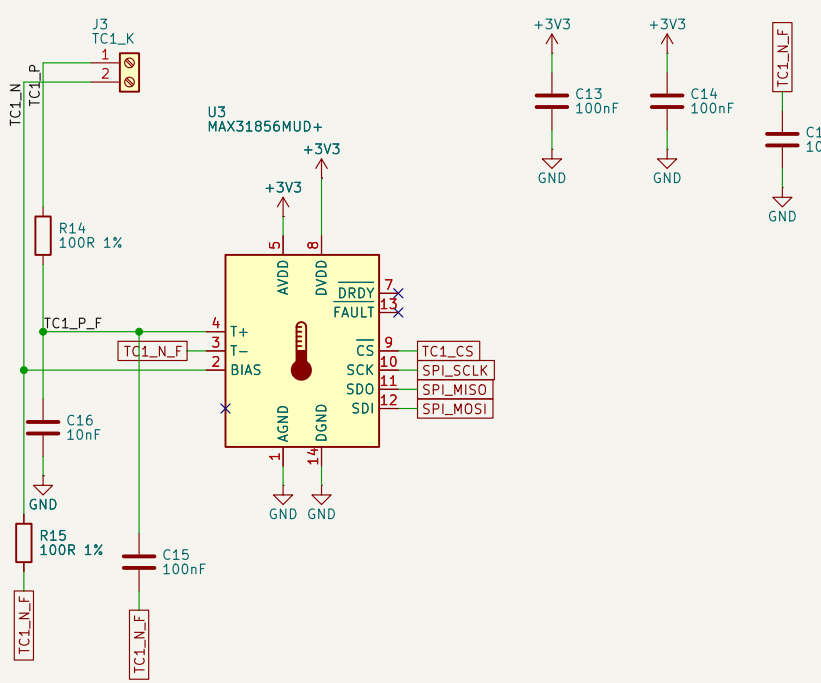
WS2812B STATUS LED
VDD dropped -4.6V for 3.3V data margin



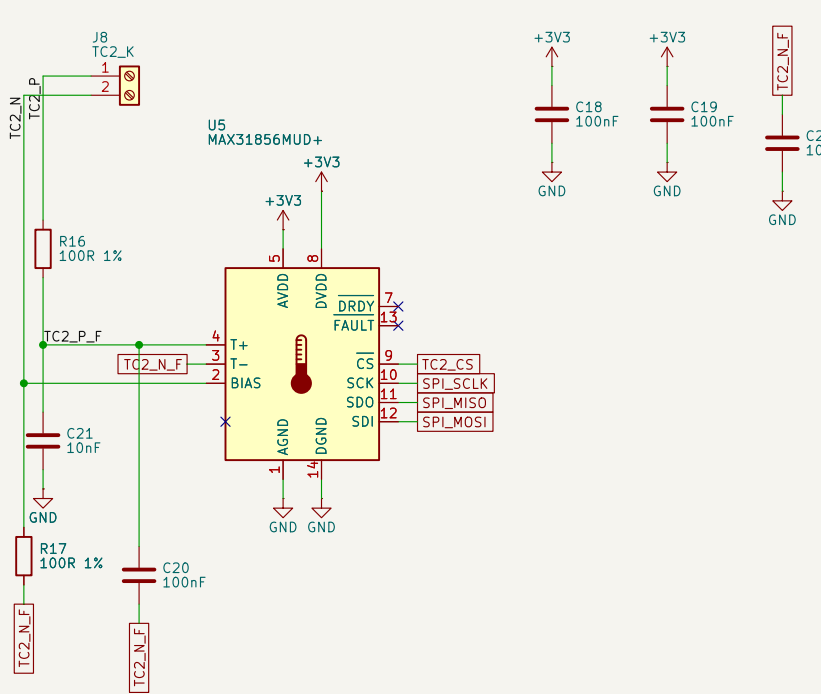
HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



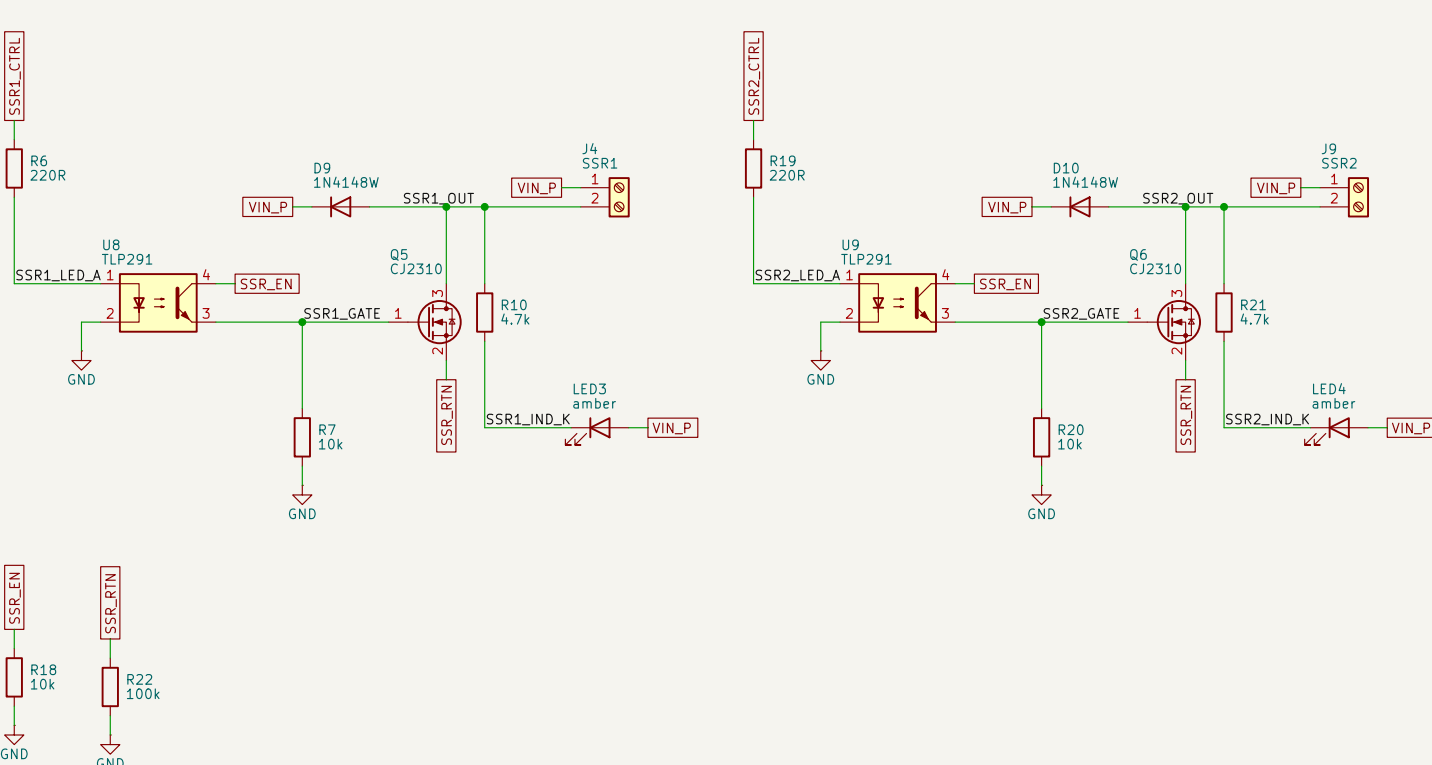
THERMOCOUPLE 1 (control) MAX31856
I- floats to J3, biased via BIAS



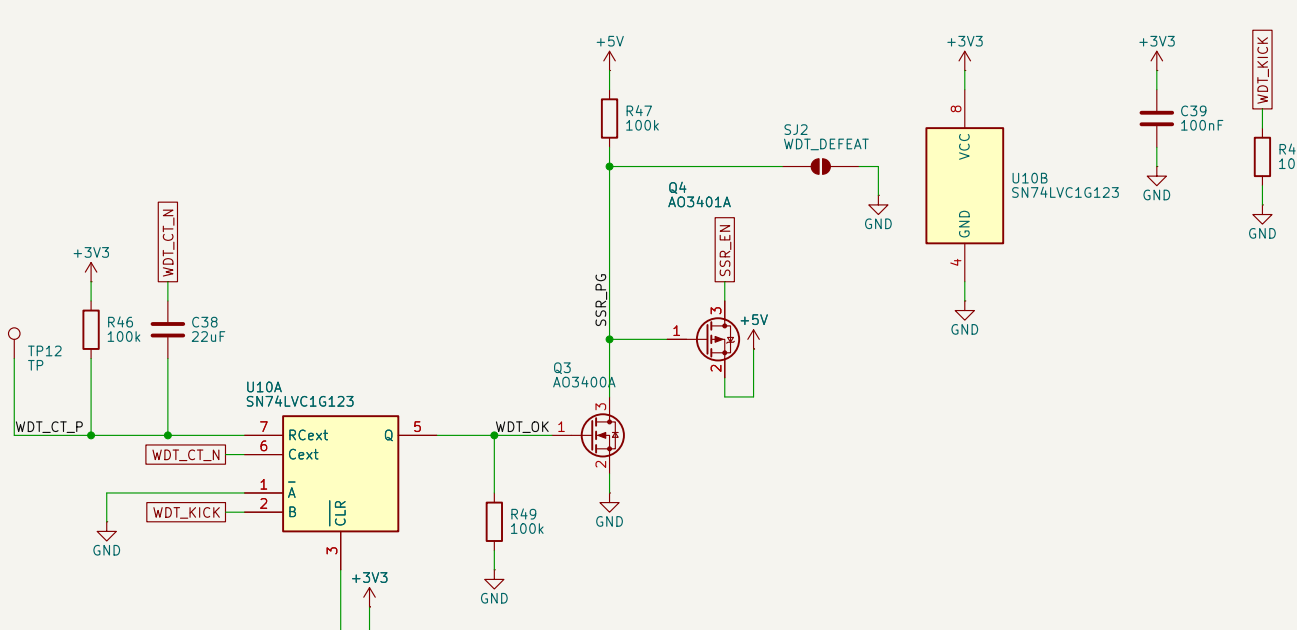
THERMOCOUPLE 2 (load) MAX31856
I- floats to J8, biased via BIAS



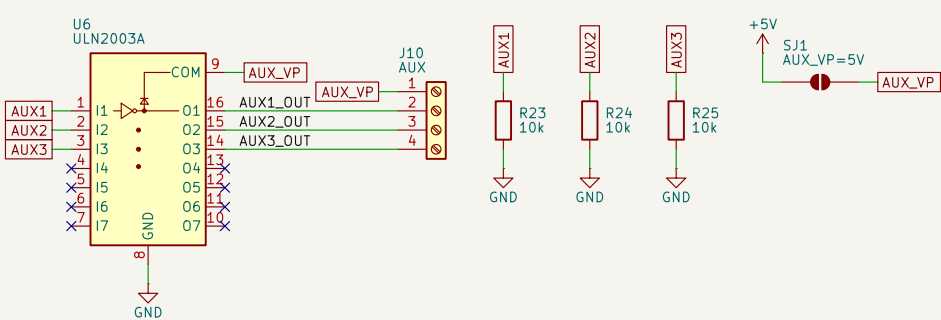
SSR DRIVE x2
24 V terminals off VIN_P, switched low side by a 60 V CJ2310. The GPIO drives an opto LED, never the gate: at 24 V a gate-to-drain failure would otherwise put 240 mA into the pin. Opto collectors sit on SSR_EN (5 V), so the gate lands at -4.9 V from the RAIL, rather than from CTR. Q7 is the watchdog's gate in the shared return - it opens a channel whose MOSFET failed short. R18/R22 define SSR_EN and SSR_RTN when their switches are off. NOT an isolation barrier - see design.py's SSR block.



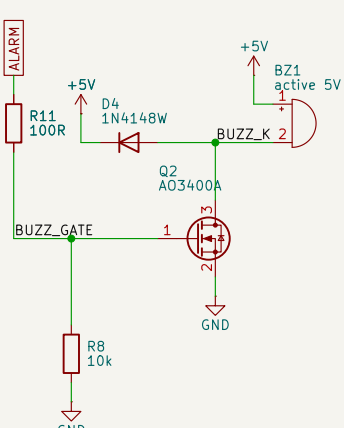
HARDWARE WATCHDOG
U10 retriggerable one-shot: every rising edge on WDT_KICK restarts a ~2.2 s window, so Q stays high only while firmware keeps kicking. Q holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN - the +5V GATE DRIVE rail feeding both opto collectors and Q7's gate.
R46/C38 set the window. R48 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up. SJ2 = bring-up defeat, REMOVE for service.



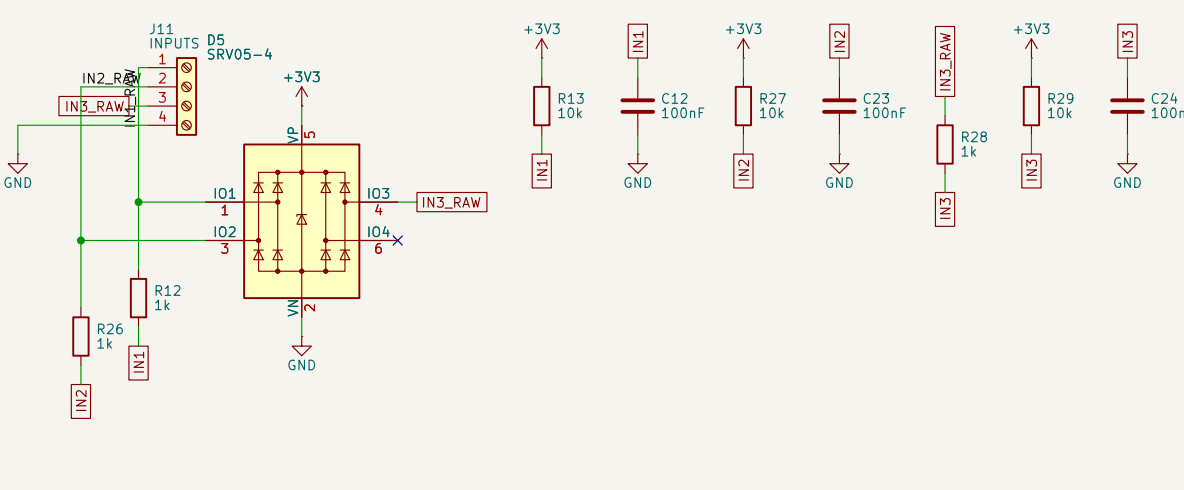
AUX OUTPUT BANK
ULN2003A vent/purge/spare; COM->AUX_VP, SJ1 links to +5V



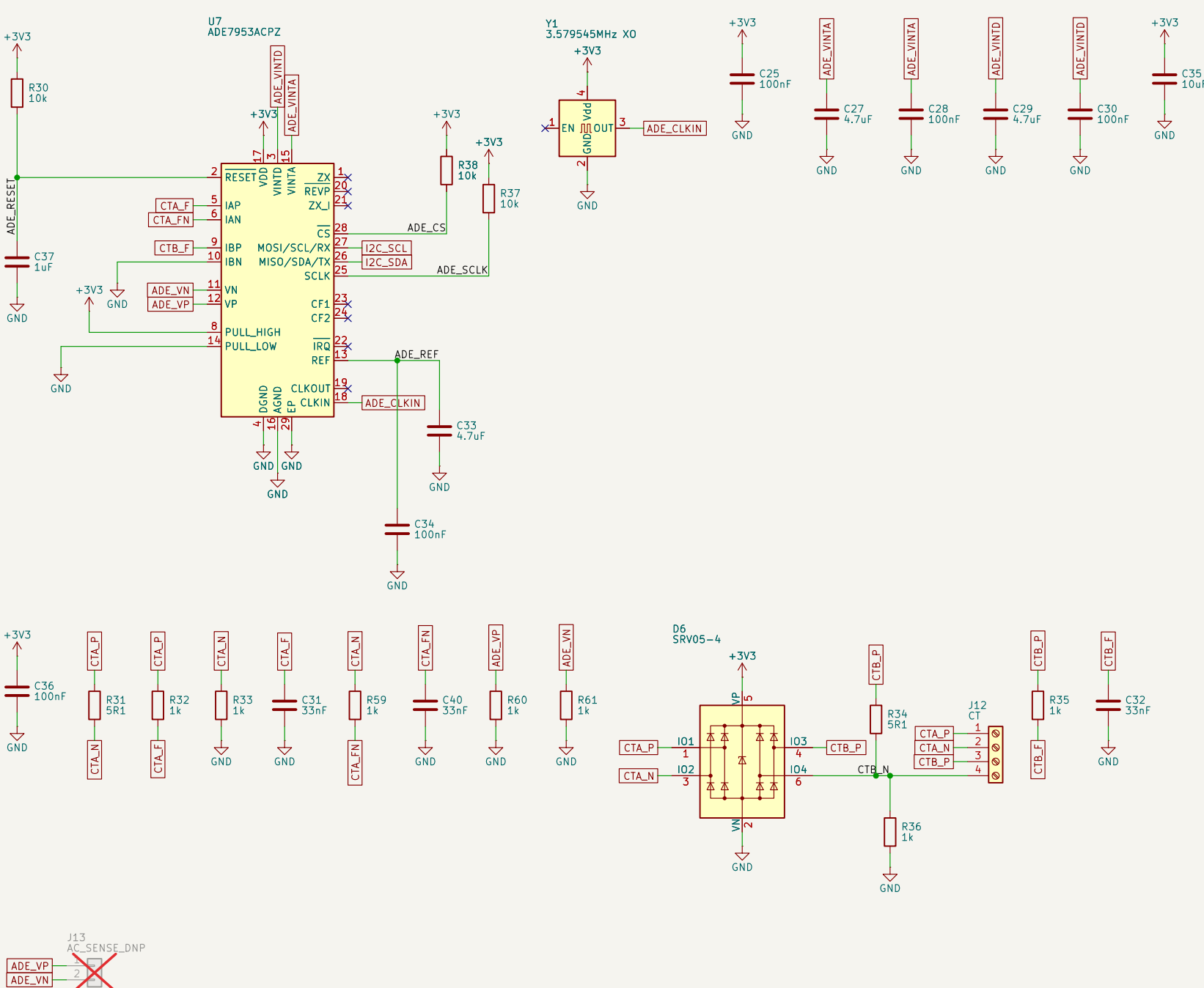
ALARM BUZZER



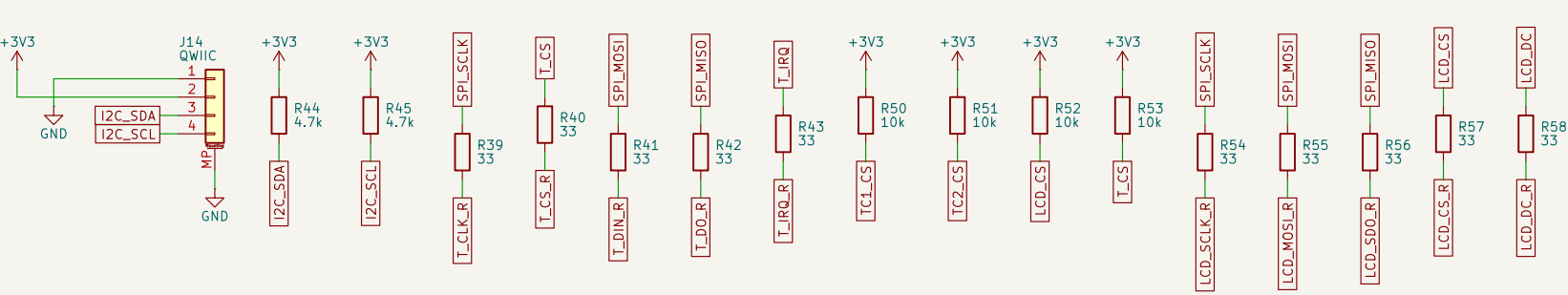
PROTECTED INPUTS x3 Iid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVs



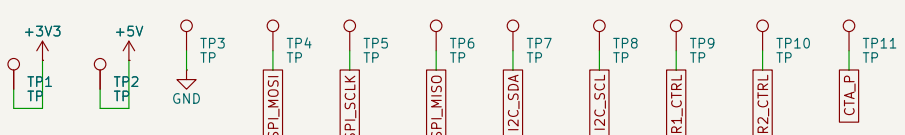
CT CURRENT SENSING ADE7953, I2C, current-only
no mains - VP/VN to DNP J13



SPI BUS DAMPING + CHIP-SELECT PULL-UPS + I2C EXPANSION
R39-R43 damp the display module's XPT2046 touch lines and R54-R58 the four 40 MHz display lines; both loom to J5. R56 is also the SDO hedge - lift it if the panel does not tri-state J5.9 off the shared SP1_MISO.
R50-R53 hold all four chip selects deasserted from power-on, before firmware configures any GPIO.
J14 Qwic and J7 5-8 (0.1 in) share I2C, pulled up by R44/R45



TEST POINTS



NOTES - Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = VIN_P (24 V, fused and TVS-clamped, ALWAYS LIVE), pin2 = the switched low side. The watchdog gates the shared return (Q7), not the supply, so a tripped watchdog leaves 24 V on pin 1, and opens pin 2. Boot-safe: R7/R20 hold each gate down while the optos are dark.
The optos are NOT an isolation barrier - both sides share GND. Rev B's opto-ISOLATED design was reverted and stays reverted: U8/U9 are here to keep a 24 V gate fault off the GND. Per channel <=200 mA; F1 holds at 750 mA and U11 takes ~250.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted: Inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (Iid/gas-flow/spare) + GND - dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RS 5=DC 6=SDI/MOSI 7=SDO 8=BL 9=SDV/MISO 10=TC1 11=TC2 12=I2C 13=I2C 14=I2C 15=I2C 16=I2C 17=I2C 18=I2C 19=I2C 20=I2C 21=I2C 22=I2C 23=I2C 24=I2C 25=I2C 26=I2C 27=I2C 28=I2C 29=I2C 30=I2C 31=I2C 32=I2C 33=I2C 34=I2C 35=I2C 36=I2C 37=I2C 38=I2C 39=I2C 40=I2C 41=I2C 42=I2C 43=I2C 44=I2C 45=I2C 46=I2C 47=I2C 48=I2C 49=I2C 50=I2C 51=I2C 52=I2C 53=I2C 54=I2C 55=I2C 56=I2C 57=I2C 58=I2C 59=I2C 60=I2C 61=I2C 62=I2C 63=I2C 64=I2C 65=I2C 66=I2C 67=I2C 68=I2C 69=I2C 70=I2C 71=I2C 72=I2C 73=I2C 74=I2C 75=I2C 76=I2C 77=I2C 78=I2C 79=I2C 80=I2C 81=I2C 82=I2C 83=I2C 84=I2C 85=I2C 86=I2C 87=I2C 88=I2C 89=I2C 90=I2C 91=I2C 92=I2C 93=I2C 94=I2C 95=I2C 96=I2C 97=I2C 98=I2C 99=I2C 100=I2C 101=I2C 102=I2C 103=I2C 104=I2C 105=I2C 106=I2C 107=I2C 108=I2C 109=I2C 110=I2C 111=I2C 112=I2C 113=I2C 114=I2C 115=I2C 116=I2C 117=I2C 118=I2C 119=I2C 120=I2C 121=I2C 122=I2C 123=I2C 124=I2C 125=I2C 126=I2C 127=I2C 128=I2C 129=I2C 130=I2C 131=I2C 132=I2C 133=I2C 134=I2C 135=I2C 136=I2C 137=I2C 138=I2C 139=I2C 140=I2C 141=I2C 142=I2C 143=I2C 144=I2C 145=I2C 146=I2C 147=I2C 148=I2C 149=I2C 150=I2C 151=I2C 152=I2C 153=I2C 154=I2C 155=I2C 156=I2C 157=I2C 158=I2C 159=I2C 160=I2C 161=I2C 162=I2C 163=I2C 164=I2C 165=I2C 166=I2C 167=I2C 168=I2C 169=I2C 170=I2C 171=I2C 172=I2C 173=I2C 174=I2C 175=I2C 176=I2C 177=I2C 178=I2C 179=I2C 180=I2C 181=I2C 182=I2C 183=I2C 184=I2C 185=I2C 186=I2C 187=I2C 188=I2C 189=I2C 190=I2C 191=I2C 192=I2C 193=I2C 194=I2C 195=I2C 196=I2C 197=I2C 198=I2C 199=I2C 200=I2C 201=I2C 202=I2C 203=I2C 204=I2C 205=I2C 206=I2C 207=I2C 208=I2C 209=I2C 210=I2C 211=I2C 212=I2C 213=I2C 214=I2C 215=I2C 216=I2C 217=I2C 218=I2C 219=I2C 220=I2C 221=I2C 222=I2C 223=I2C 224=I2C 225=I2C 226=I2C 227=I2C 228=I2C 229=I2C 230=I2C 231=I2C 232=I2C 233=I2C 234=I2C 235=I2C 236=I2C 237=I2C 238=I2C 239=I2C 240=I2C 241=I2C 242=I2C 243=I2C 244=I2C 245=I2C 246=I2C 247=I2C 248=I2C 249=I2C 250=I2C 251=I2C 252=I2C 253=I2C 254=I2C 255=I2C 256=I2C 257=I2C 258=I2C 259=I2C 260=I2C 261=I2C 262=I2C 263=I2C 264=I2C 265=I2C 266=I2C 267=I2C 268=I2C 269=I2C 270=I2C 271=I2C 272=I2C 273=I2C 274=I2C 275=I2C 276=I2C 277=I2C 278=I2C 279=I2C 280=I2C 281=I2C 282=I2C 283=I2C 284=I2C 285=I2C 286=I2C 287=I2C 288=I2C 289=I2C 290=I2C 291=I2C 292=I2C 293=I2C 294=I2C 295=I2C 296=I2C 297=I2C 298=I2C 299=I2C 300=I2C 301=I2C 302=I2C 303=I2C 304=I2C 305=I2C 306=I2C 307=I2C 308=I2C 309=I2C 310=I2C 311=I2C 312=I2C 313=I2C 314=I2C 315=I2C 316=I2C 317=I2C 318=I2C 319=I2C 320=I2C 321=I2C 322=I2C 323=I2C 324=I2C 325=I2C 326=I2C 327=I2C 328=I2C 329=I2C 330=I2C 331=I2C 332=I2C 333=I2C 334=I2C 335=I2C 336=I2C 337=I2C 338=I2C 339=I2C 340=I2C 341=I2C 342=I2C 343=I2C 344=I2C 345=I2C 346=I2C 347=I2C 348=I2C 349=I2C 350=I2C 351=I2C 352=I2C 353=I2C 354=I2C 355=I2C 356=I2C 357=I2C 358=I2C 359=I2C 360=I2C 361=I2C 362=I2C 363=I2C 364=I2C 365=I2C 366=I2C 367=I2C 368=I2C 369=I2C 370=I2C 371=I2C 372=I2C 373=I2C 374=I2C 375=I2C 376=I2C 377=I2C 378=I2C 379=I2C 380=I2C 381=I2C 382=I2C 383=I2C 384=I2C 385=I2C 386=I2C 387=I2C 388=I2C 389=I2C 390=I2C 391=I2C 392=I2C 393=I2C 394=I2C 395=I2C 396=I2C 397=I2C 398=I2C 399=I2C 400=I2C 401=I2C 402=I2C 403=I2C 404=I2C 405=I2C 406=I2C 407=I2C 408=I2C 409=I2C 410=I2C 411=I2C 412=I2C 413=I2C 414=I2C 415=I2C 416=I2C 417=I2C 418=I2C 419=I2C 420=I2C 421=I2C 422=I2C 423=I2C 424=I2C 425=I2C 426=I2C 427=I2C 428=I2C 429=I2C 430=I2C 431=I2C 432=I2C 433=I2C 434=I2C 435=I2C 436=I2C 437=I2C 438=I2C 439=I2C 440=I2C 441=I2C 442=I2C 443=I2C 444=I2C 445=I2C 446=I2C 447=I2C 448=I2C 449=I2C 450=I2C 451=I2C 452=I2C 453=I2C 454=I2C 455=I2C 456=I2C 457=I2C 458=I2C 459=I2C 460=I2C 461=I2C 462=I2C 463=I2C 464=I2C 465=I2C 466=I2C 467=I2C 468=I2C 469=I2C 470=I2C 471=I2C 472=I2C 473=I2C 474=I2C 475=I2C 476=I2C 477=I2C 478=I2C 479=I2C 480=I2C 481=I2C 482=I2C 483=I2C 484=I2C 485=I2C 486=I2C 487=I2C 488=I2C 489=I2C 490=I2C 491=I2C 492=I2C 493=I2C 494=I2C 495=I2C 496=I2C 497=I2C 498=I2C 499=I2C 500=I2C 501=I2C 502=I2C 503=I2C 504=I2C 505=I2C 506=I2C 507=I2C 508=I2C 509=I2C 510=I2C 511=I2C 512=I2C 513=I2C 514=I2C 515=I2C 516=I2C 517=I2C 518=I2C 519=I2C 520=I2C 521=I2C 522=I2C 523=I2C 524=I2C 525=I2C 526=I2C 527=I2C 528=I2C 529=I2C 530=I2C 531=I2C 532=I2C 533=I2C 534=I2C 535=I2C 536=I2C 537=I2C 538=I2C 539=I2C 540=I2C 541=I2C 542=I2C 543=I2C 544=I2C 545=I2C 546=I2C 547=I2C 548=I2C 549=I2C 550=I2C 551=I2C 552=I2C 553=I2C 554=I2C 555=I2C 556=I2C 557=I2C 558=I2C 559=I2C 560=I2C 561=I2C 562=I2C 563=I2C 564=I2C 565=I2C 566=I2C 567=I2C 568=I2C 569=I2C 570=I2C 571=I2C 572=I2C 573=I2C 574=I2C 575=I2C 576=I2C 577=I2C 578=I2C 579=I2C 580=I2C 581=I2C 582=I2C 583=I2C 584=I2C 585=I2C 586=I2C 587=I2C 588=I2C 589=I2C 590=I2C 591=I2C 592=I2C 593=I2C 594=I2C 595=I2C 596=I2C 597=I2C 598=I2C 599=I2C 600=I2C 601=I2C 602=I2C 603=I2C 604=I2C 605=I2C 606=I2C 607=I2C 608=I2C 609=I2C 610=I2C 611=I2C 612=I2C 613=I2C 614=I2C 615=I2C 616=I2C 617=I2C 618=I2C 619=I2C 620=I2C 621=I2C 622=I2C 623=I2C 624=I2C 625=I2C 626=I2C 627=I2C 628=I2C 629=I2C 630=I2C 631=I2C 632=I2C 633=I2C 634=I2C 635=I2C 636=I2C 637=I2C 638=I2C 639=I2C 640=I2C 641=I2C 642=I2C 643=I2C 644=I2C 645=I2C 646=I2C 647=I2C 648=I2C 649=I2C 650=I2C 651=I2C 652=I2C 653=I2C 654=I2C 655=I2C 656=I2C 657=I2C 658=I2C 659=I2C 660=I2C 661=I2C 662=I2C 663=I2C 664=I2C 665=I2C 666=I2C 667=I2C 668=I2C 669=I2C 670=I2C 671=I2C 672=I2C 673=I2C 674=I2C 675=I2C 676=I2C 677=I2C 678=I2C 679=I2C 680=I2C 681=I2C 682=I2C 683=I2C 684=I2C 685=I2C 686=I2C 687=I2C 688=I2C 689=I2C 690=I2C 691=I2C 692=I2C 693=I2C 694=I2C 695=I2C 696=I2C 697=I2C 698=I2C 699=I2C 700=I2C 701=I2C 702=I2C 703=I2C 704=I2C 705=I2C 706=I2C 707=I2C 708=I2C 709=I2C 710=I2C 711=I2C 712=I2C 713=I2C 714=I2C 715=I2C 716=I2C 717=I2C 718=I2C 719=I2C 720=I2C 721=I2C 722=I2C 723=I2C 724=I2C 725=I2C 726=I2C 727=I2C 728=I2C 729=I2C 730=I2C 731=I2C 732=I2C 733=I2C 734=I2C 735=I2C 736=I2C 737=I2C 738=I2C 739=I2C 740=I2C 741=I2C 742=I2C 743=I2C 744=I2C 745=I2C 746=I2C 747=I2C 748=I2C 749=I2C 750=I2C 751=I2C 752=I2C 753=I2C 754=I2C 755=I2C 756=I2C 757=I2C 758=I2C 759=I2C 760=I2C 761=I2C 762=I2C 763=I2C 764=I2C 765=I2C 766=I2C 767=I2C 768=I2C 769=I2C 770=I2C 771=I2C 772=I2C 773=I2C 774=I2C 775=I2C 776=I2C 777=I2C 778=I2C 779=I2C 780=I2C 781=I2C 782=I2C 783=I2C 784=I2C 785=I2C 786=I2C 787=I2C 788=I2C 789=I2C 790=I2C 791=I2C 792=I2C 793=I2C 794=I2C 795=I2C 796=I2C 797=I2C 798=I2C 799=I2C 800=I2C 801=I2C 802=I2C 803=I2C 804=I2C 805=I2C 806=I2C 807=I2C 808=I2C 809=I2C 810=I2C 811=I2C 812=I2C 813=I2C 814=I2C 815=I2C 816=I2C 817=I2C 818=I2C 819=I2C 820=I2C 821=I2C 822=I2C 823=I2C 824=I2C 825=I2C 826=I2C 827=I2C 828=I2C 829=I2C 830=I2C 831=I2C 832=I2C 833=I2C 834=I2C 835=I2C 836=I2C 837=I2C 838=I2C 839=I2C 840=I2C 841=I2C 842=I2C 843=I2C 844=I2C 845=I2C 846=I2C 847=I2C 848=I2C 849=I2C 850=I2C 851=I2C 852=I2C 853=I2C 854=I2C 855=I2C 856=I2C 857=I2C 858=I2C 859=I2C 860=I2C 861=I2C 862=I2C 863=I2C 864=I2C 865=I2C 866=I2C 867=I2C 868=I2C 869=I2C 870=I2C 871=I2C 872=I2C 873=I2C 874=I2C 875=I2C 876=I2C 877=I2C 878=I2C 879=I2C 880=I2C 881=I2C 882=I2C 883=I2C 884=I2C 885=I2C 886=I2C 887=I2C 888=I2C 889=I2C 890=I2C 891=I2C 892=I2C 893=I2C 894=I2C 895=I2C 896=I2C 897=I2C 898=I2C 899=I2C 900=I2C 901=I2C 902=I2C 903=I2C 904=I2C 905=I2C 906=I2C 907=I2C 908=I2C 909=I2C 910=I2C 911=I2C 912=I2C 913=I2C 914=I2C 915=I2C 916=I2C 917=I2C 918=I2C 919=I2C 920=I2C 921=I2C 922=I2C 923=I2C 924=I2C 925=I2C 926=I2C 927=I2C 928=I2C 929=I2C 930=I2C 931=I2C 932=I2C 933=I2C 934=I2C 935=I2C 936=I2C 937=I2C 938=I2C 939=I2C 940=I2C 941=I2C 942=I2C 943=I2C 944=I2C 945=I2C 946=I2C 947=I2C 948=I2C 949=I2C 950=I2C 951=I2C 952=I2C 953=I2C 954=I2C 955=I2C 956=I2C 957=I2C 958=I2C 959=I2C 960=I2C 961=I2C 962=I2C 963=I2C 964=I2C 965=I2C 966=I2C 967=I2C 968=I2C 969=I2C 970=I2C 971=I2C 972=I2C 973=I2C 974=I2C 975=I2C 976=I2C 977=I2C 978=I2C 979=I2C 980=I2C 981=I2C 982=I2C 983=I2C 984=I2C 985=I2C 986=I2C 987=I2C 988=I2C 989=I2C 990=I2C 991=I2C 992=I2C 993=I2C 994=I2C 995=I2C 996=I2C 997=I2C 998=I2C 999=I2C 1000=I2C 1001=I2C 1002=I2C 1003=I2C 1004=I2C 1005=I2C 1006=I2C 1007=I2C 1008=I2C 1009=I2C 1010=I2C 1011=I2C 1012=I2C 1013=I2C 1014=I2C 1015=I2C 1016=I2C 1017=I2C 1018=I2C 1019=I2C 1020=I2C 1021=I2C 1022=I2C 1023=I2C 1024=I2C 1025=I2C 1026=I2C 1027=I2C 1028=I2C 1029=I2C 1030=I2C 1031=I2C 1032=I2C 1033=I2C 1034=I2C 1035=I2C 1036=I2C 1037=I2C 1038=I2C 1039=I2C 1040=I2C 1041=I2C 1042=I2C 1043=I2C 1044=I2C 1045=I2C 1046=I2C 1047=I2C 1048=I2C 1049=I2C 1050=I2C 1051=I2C 1052=I2C 1053=I2C 1054=I2C 1055=I2C 1056=I2C 1057=I2C 1058=I2C 1059=I2C 1060=I2C 1061=I2C 1062=I2C 1063=I2C 1064=I2C 1065=I2C 1066=I2C 1067=I2C 1068=I2C 1069=I2C 1070=I2C 1071=I2C 1072=I2C 1073=I2C 1074=I2C 1075=I2C 1076=I2C 1077=I2C 1078=I2C 1079=I2C 1080=I2C 1081=I2C 1082=I2C 1083=I2C 1084=I2C 1085=I2C 1086=I2C 1087=I2C 1088=I2C 1089=I2C 1090=I2C 1091=I2C 1092=I2C 1093=I2C 1094=I2C 1095=I2C 1096=I2C 1097=I2C 1098=I2C 1099=I2C 1100=I2C 1101=I2C 1102=I2C 1